

Can a positive average return be just luck?

Week 3 sample lecture: Hypothesis Testing of Stock Returns

+2.31% average monthly return

Forty-eight observations. Impressive, or statistically noisy?

CLASS VOTE

Before we calculate anything: **would you invest based on this average alone?**

A. Yes B. No C. Not enough information

LEARNING OBJECTIVE

By the end of this mini-lecture, you can test whether a stock's mean return differs from zero using a one-sample t-test, and explain what the result does **and does not** mean for an investor.

Step 1: Turn the investment claim into a test

We begin with two competing statements.

NULL HYPOTHESIS

H0: mean return = 0

The observed average can be explained by sampling noise around a zero mean.

ALTERNATIVE HYPOTHESIS

H1: mean return is not 0

The underlying mean return is different from zero.

Key idea

A positive sample average is not automatically evidence of a positive population mean. We ask how unusual our sample would be if H0 were true.

Step 2: Measure signal relative to noise

The one-sample t-statistic

$$t = \text{signal} / \text{standard error}$$

How far the mean is from zero, scaled by sampling uncertainty

SIGNAL

Sample mean - benchmark.

How far the observed average sits from the null value.

NOISE

Standard deviation / $\sqrt{n}$.

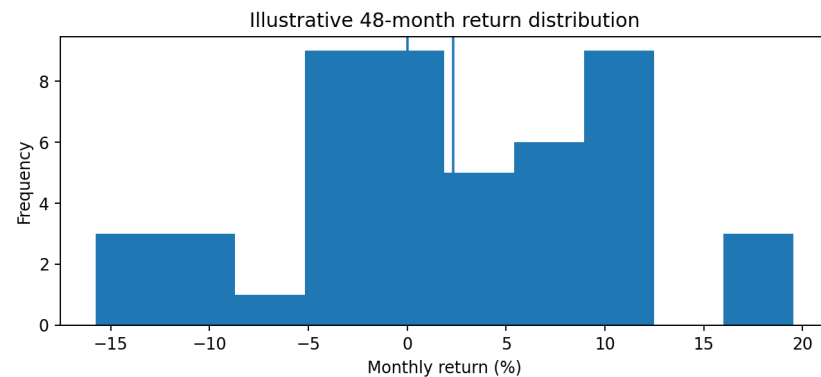
How much random sampling variation we expect.

DECISION

Larger absolute t means the signal is stronger relative to the noise.

The case: 48 months of illustrative stock returns

Same mean. But now we can see the noise.



OBSERVED

n = 48
mean = **2.31%**
std. dev. = **8.42%**

QUESTION

Is +2.31% large enough
relative to 8.42%
volatility?

Step 3: Calculate the test statistic

Now the average has context.

$$t = (2.31 - 0) / (8.42 / \text{sqrt}(48))$$

$$t = 1.90$$

INTERPRETATION

The sample mean is about **1.90 standard errors** above zero.

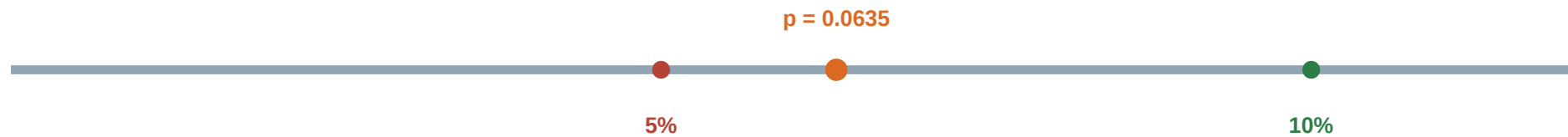
TWO-TAILED P-VALUE

$$p = 0.0635$$

How surprising is a result this extreme if the true mean were zero?

The reveal: suggestive, but not strong enough at 5%

Statistical significance depends on the decision threshold.



AT ALPHA = 0.05

$p > 0.05 \rightarrow$ **fail to reject H_0 .**

The evidence is not strong enough for the conventional 5% threshold.

AT ALPHA = 0.10

$p < 0.10 \rightarrow$ **reject H_0 .**

Evidence is suggestive, but the conclusion is sensitive to the chosen threshold.

Your turn: test a second stock

15-minute paired exercise

STUDENT TASK

Using a second 48-month return series:

1. Compute the sample mean and standard deviation.
2. Test H_0 : mean return = 0.
3. Report t and p .
4. Decide at $\alpha = 0.05$ and $\alpha = 0.10$.
5. Write one sentence for an investor: **what does the result actually tell us?**

EXIT QUESTION

Why is ' $p = 0.06$ ' not the same as 'there is a 6% chance H_0 is true'?

Explain in plain English.

From formula -> evidence -> decision.